

- 4 A progressive wave from a source moves past two points P and Q.

Fig. 4.1 shows the variation with time t of the displacement y at P.

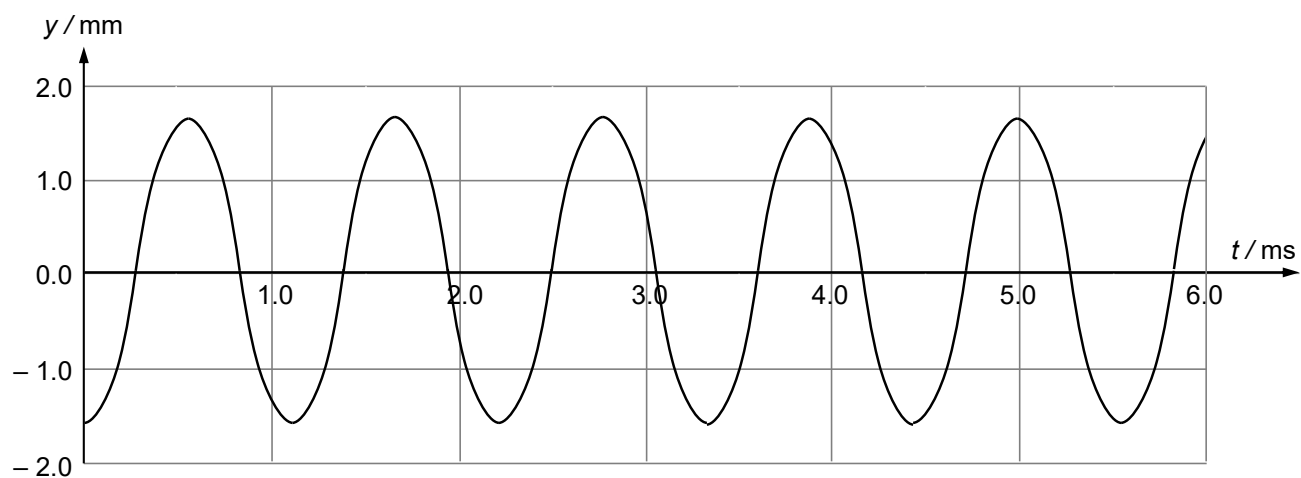


Fig. 4.1

Fig. 4.2 shows how the displacement of the wave at time $t = 0$ varies with distance x from the source.

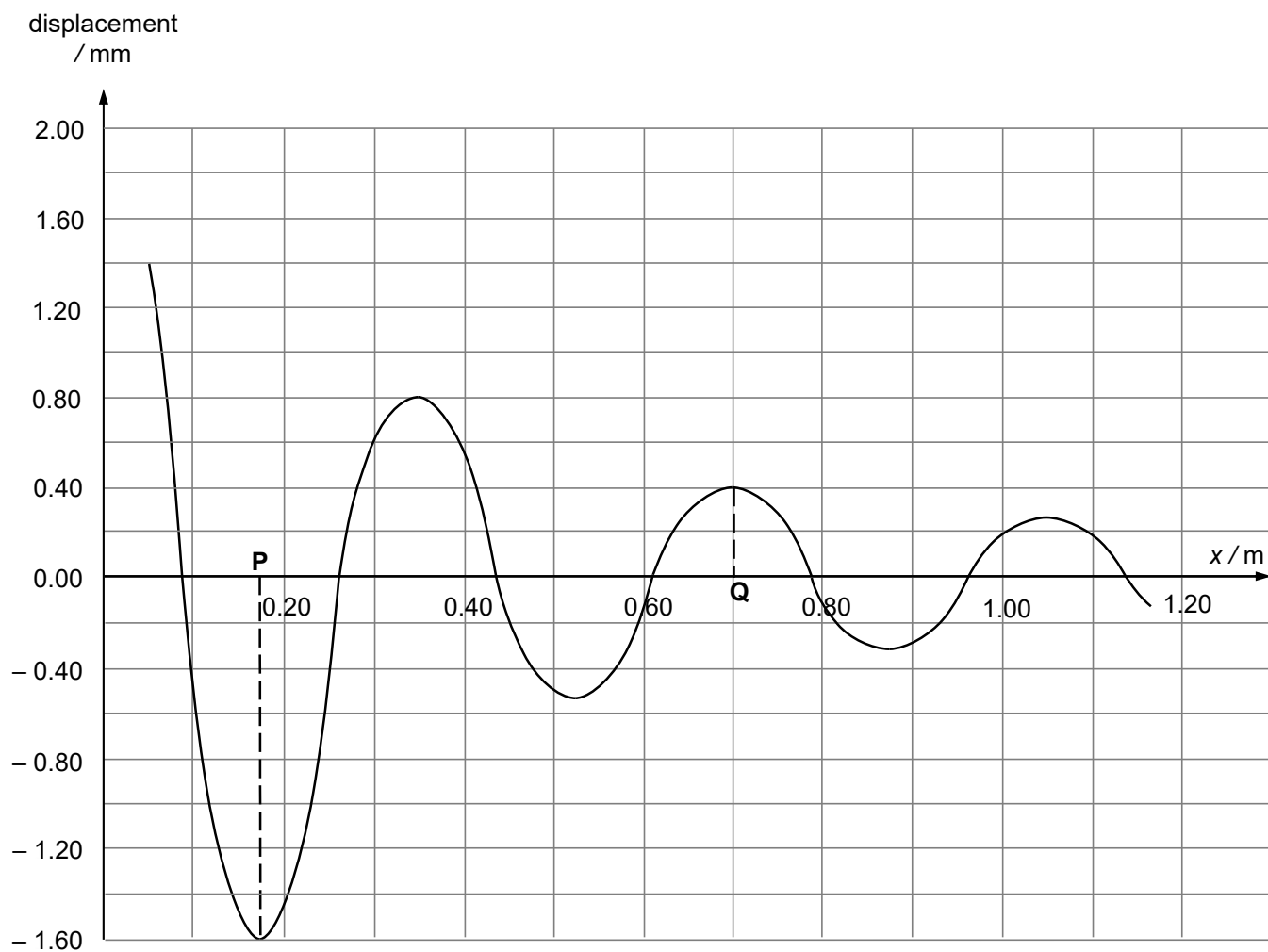


Fig. 4.2

- (a) Explain what is meant by the term *progressive* as applied to a wave.

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- (b) Using data from Fig. 4.1 and 4.2, deduce for this wave

- (i) the wavelength,

wavelength = m [1]

- (ii) the frequency,

frequency = Hz [1]

- (iii) the speed.

speed = m s^{-1} [1]

- (c) The intensity of the wave is inversely proportional to the square of the distance from the source.

Without drawing a graph, use the data from Fig. 4.1 and 4.2 to determine whether the wave source in (b) obeys this relationship.

conclusion[3]

[Total: 8]